Bilaga B

VisCon - Course material

B.1 How to apply Viscon

B.1.1 Method

1. Define model:

The initial step is to update VisCon with model data for the current problem, i.e. insert values for given soil profile and load case. Furthermore, when defining the model it is important to choose an element and time increment size small enough to achieve a converging solution.

2. Extract 1D-plot:

Next step is to define the maximum time for the time simulation, and make sure that the program record values of the excess pore water pressure at desired time steps. This is done by choosing a time increment dividable with desired length of time steps, and a number of screenshots catching these, see the example in B.1.2. Furthermore, the user should define intervals for the vertical and horisontal axis where one dimensional plots of the excess pore water pressure should be extracted.

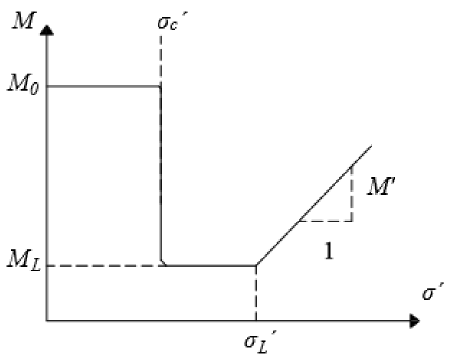
3. Calculate settlements:

The process of consolidation results in settlements. Settlements depend on the effective stress, i.e. the pressure carried by the soil skeleton. The total settlement δ_z

can be calculated by integrating the strain function over the depth at a certain section, according to:

$$\delta_z = \int_0^h \varepsilon_z dz \tag{B.1}$$

The strain ε_z depends on the consolidation level and is described by either Equation B.2, B.3 or B.4. $\sigma'_{z,0}$ is the effective stress before an increased stress and $\sigma'_{z,1}$ is the effective stress after an increased stress, i.e. $\sigma'_{z,1} = \sigma'_{z,0} + \Delta\sigma'_z$. Figure B.1 shows the relation between the compression modulus M and the effective stress σ'. σ'_c is referred to as the consolidation stress and σ'_L is the effective stress for which the compression modulus M is no longer constant.



Figur B.1: *Compression modulus M as a function of the effective stress σ' for clay.*

If $\sigma'_{z,1} \leq \sigma'_c$ the strain is given by:

$$\varepsilon_z = \frac{\Delta\sigma'_z}{M_0} \tag{B.2}$$

If $\sigma'_c < \sigma'_{z,1} \leq \sigma'_L$ the strain is given by:

$$\varepsilon_z = \frac{\sigma'_c - \sigma'_{z,0}}{M_0} + \frac{\sigma'_{z,1} - \sigma'_c}{M_L} \tag{B.3}$$

If $\sigma'_{z,1} > \sigma'_L$ the strain is given by:

$$\varepsilon_z = \frac{\sigma'_c - \sigma'_{z,0}}{M_0} + \frac{\sigma'_L - \sigma'_c}{M_L} + \frac{1}{M'} \ln\left(1 + (\sigma'_{z,1} - \sigma'_L)\frac{M'}{M_L}\right) \tag{B.4}$$

When calculating the total settlement it is common to divide the soil into a number of layers. For each layer the average effective stress is calculated and multiplied by the layer depth. If the soil profile is divided into n layers the total settlement can be expressed as a sum, according to:

$$\delta_z = \sum_{i=1}^{n} \varepsilon_{z,i} h_i \tag{B.5}$$

B.1.2 Example

Problem

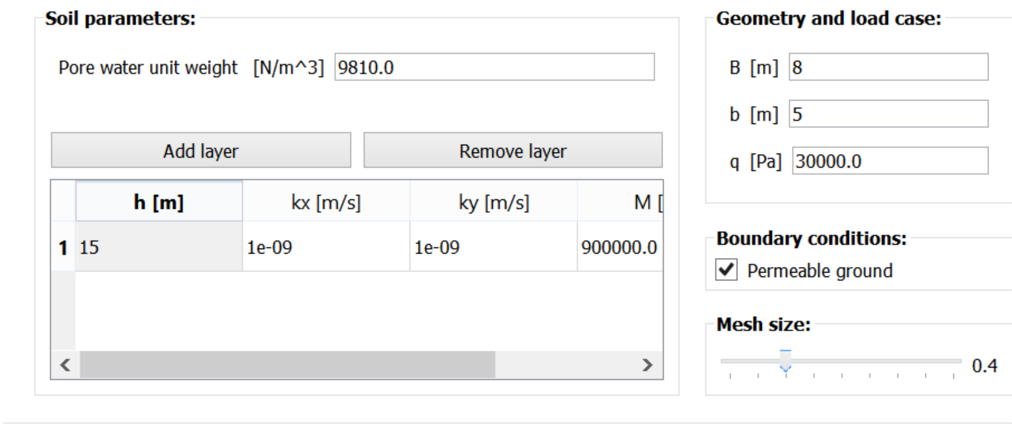
Consider a 15 m thick layer of clay resting on sand. The clay material has the permeability $k_x = k_z = 10^{-9}$ m/s and the compression modulus $M_L = 900$ kPa. Furthermore $\sigma'_{z,0} = \sigma'_c$. A road embankment is built on top of this layer. It has a width of 8 meters and is estimated to contribute with an increased load of 30 kPa. Calculate the settlement after 2 years and after a long time for a vertical section in the centre line as well as a vertical section 6 meter from the centre line. Do this by dividing the clay into five calculation layers.

1. Define model:

A 15 m layer is created in VisCon. For this layer the permeability is set to $k_x = k_z = 10^{-9}$ m/s and the compression modulus $M_L = 900$ kPa. The width of the load, with reference to the centre line, is $b = 5$ m. The width of the model is set to $B = 8$ m. The boundary conditions are set by ticking the box Permeable ground as the sand layer underneath the clay is considered permeable. See Figure B.2 for model input data. The element size is choosen as 0.5 and the time increment as $\Delta t = 0.05$ years.

2. Extract 1D -plot:

In order to catch the time step $t = 2$ years the maximum time is set to $T = 5$ years and the number of screenshots taken is set to 5. This will produce a a slider where the two dimensional and one dimensional visualisations can be shown for $t = 0$, 1, 2, 3, 4, 5 years. The value for the number of intervals along the x-axis is set to 5 in order to be able to extract a vertical 1D-plot at $x = 0$, 2, 4, 6, 8 m.



Figur B.2: *Input data when defining the model in example exercise.*

Tabell B.1: Computed effective stress $\Delta\sigma_z'$ after 2 years.

Depth z	$x = 0$ m	$x = 6$ m
1.5 m	25.6 kPa	4.2 kPa
4.5 m	22.0 kPa	9.2 kPa
7.5 m	17.3 kPa	9.7 kPa
10.5 m	13.7 kPa	9.3 kPa
13.5 m	11.0 kPa	8.5 kPa

When the time simulation is done the one dimensional plots will be available, by choosing values of the current time step and the x-coordinate for a selected section. The first assignment is to determine the settlements at $t = 2$ years for a vertical section at $x = 0$ m and $x = 6$ m. The time slider is set at 2 and the x-value slider at $x = 0$ m respective $x = 6$ m. The figure showing the effective stress is used and Table B.1 shows the values for $\Delta\sigma_z'$ after 2 years. It is possible to zoom and pan in the figure in order to determine values easier.

As the excess pore water pressure is assumed to be zero after a long time the effective stress then equals the initial stress distrubution $\Delta\sigma_z$. Hence, the values for the effective stress after a long time is given by studying the initial excess pore water pressure. The first figure is studied for a section at $x = 0$ m and $x = 6$ m. The values are given in Table B.2.

3. Calculate settlements:

As $\sigma_{z,0}' = \sigma_c'$ Equation B.2, in combination with Equation B.5, can be simplified and written as:

Tabell B.2: Computed effective stress $\Delta\sigma'_z$ after a long time.

Depth z	$x = 0$ m	$x = 6$ m
1.5 m	29.6 kPa	5.0 kPa
4.5 m	25.5 kPa	10.5 kPa
7.5 m	20.0 kPa	11.2 kPa
10.5 m	16.0 kPa	10.7 kPa
13.5 m	13.0 kPa	9.8 kPa

$$\delta = \sum_{i=1}^{n} \frac{\Delta\sigma'_{z,i}}{M_L} h_i \tag{B.6}$$

For $t = 2$ years and $x = 0$ m the following settlement is calculated:

$$\delta_{t=2,x=0} = \frac{25.6}{900}3.0 + \frac{22.0}{900}3.0 + \frac{17.3}{900}3.0 + \frac{13.7}{900}3.0 + \frac{11.0}{900}3.0 = 0.30 \text{ m}$$

For $t = 2$ years and $x = 6$ m the following settlement is calculated:

$$\delta_{t=2,x=6} = \frac{4.2}{900}3.0 + \frac{9.2}{900}3.0 + \frac{9.7}{900}3.0 + \frac{9.3}{900}3.0 + \frac{8.5}{900}3.0 = 0.14 \text{ m}$$

For $t = \infty$ years and $x = 0$ m the following settlement is calculated:

$$\delta_{t=\infty,x=0} = \frac{29.6}{900}3.0 + \frac{25.5}{900}3.0 + \frac{20.0}{900}3.0 + \frac{16.0}{900}3.0 + \frac{13.0}{900}3.0 = 0.35 \text{ m}$$

For $t = \infty$ years and $x = 6$ m the following settlement is calculated:

$$\delta_{t=\infty,x=6} = \frac{5.0}{900}3.0 + \frac{10.5}{900}3.0 + \frac{11.2}{900}3.0 + \frac{10.7}{900}3.0 + \frac{9.8}{900}3.0 = 0.16 \text{ m}$$

B.2 Exercises

A1

Consider a 15 m deep layer of clay, resting on sand. The clay has the material parameters $\sigma'_{z,0} = \sigma'_c$, $k_x = k_z = 5{\cdot}10^{-9}$ m/s and $M_L = 600$ kPa. A road embankment with a width of 14 m causes a surface load of 40 kPa.

a) Calculate the excess pore water pressure at the centre line after 5 years.

b) Calculate the total settlement after 5 years and after a long time. What is the avarage degree of consolidation after 5 years?

c) Compare the results in a) - b) with the results in Exercise 5-3 in *Course literature in Foundation Engineering VGTN01 2017* [21]. Make a comment upon the differences.

A2

Consider the model defined in Exercise A1. Calculate the settlements using $n = 1$, 2, 3 and 5 calculation layers. Plot the relationship between the number of layers and the calculated settlements. Make a comment upon when the solution convergens.

A3

Consider the model defined in Exercise A1, but loaded with a road embankment with the width of 2 m.

a) At the centre line, calculate the total settlement after 5 years. Compare the result with the results in Exercise A1 b) and Exercise 5-3 b) in *Course literature in Foundation Engineering VGTN01 2017* [21].

b) Is a one dimensional model appropriate for the load case described in this exercise?

c) Which are the limitations of a one dimensional versus a two dimensional model when calculating settlements?

A4

Consider the model defined in exercise A1 but with the clay layer resting on solid rock instead of sand.

a) Calculate the settlement after 5 years for a vertical section at $x = 0$ m, $x = 4$ m and $x = 8$ m.

b) Compare the result for $x = 0$ m with the result in Exercise A1 b).

c) Make a sketch of the deformed model.

A5

Try a model in VisCon with various values of time increment and element size. How will these parameters affect the solution? Furthermore, the flow, q, perpendicular to the right boundary is modelled as zero. Vary the width of the model, and discuss under which conditions this implementation is valid.